RAFFLES INSTITUTION
H2 Mathematics (9758)
2026 Year 6

Tutorial S4: Sampling

Section A (Basic Questions)

1 There were 500 spectators seated at the grandstand of a stadium. A surveyor took the seat plan of the grandstand and threw a die on the seat plan 20 times and interviewed the spectators at the seats on which the die landed. Give a reason why this might not result in a random sample of spectators.

VJC Prelim 9740/2014/02/Q5 (part)
Solution:
The surveyor is likely to avoid throwing at the edge of the map. Hence not all the 500 seats have an equal chance of being selected. <u>Alternative</u> After a seat is selected, the surveyor is likely to avoid the nearby seats. Hence the selections are not independent of one another.

2 In each of the following cases, find unbiased estimates of the population mean and population variance of X.

(a) sample size = 100, $\sum x = 160$, $\sum x^2 = 265$,

(b) sample size = 150, $\sum (x-154) = 150$, $\sum (x-154)^2 = 11000$.

[**(a)** $\frac{8}{5}$, $\frac{1}{11}$ **(b)** 155, $\frac{10850}{149}$]

Solution:	
(a)	An unbiased estimate of population mean is $\overline{x} = \frac{\sum x}{100} = \frac{8}{5} = 1.6$. An unbiased estimate of population variance is $s^2 = \frac{1}{n-1}\left[\sum x^2 - \frac{\left(\sum x\right)^2}{n}\right] = \frac{1}{99}\left(265 - \frac{(160)^2}{100}\right) = \frac{1}{11}$. **Note:** The following can be found in MF27 Unbiased estimate of population variance: $s^2 = \frac{n}{n-1}\left(\frac{\sum (x-\overline{x})^2}{n}\right) = \frac{1}{n-1}\left(\sum x^2 - \frac{\left(\sum x\right)^2}{n}\right)$

(b) Let $y = x - 154$

Then $\bar{y} = \dfrac{\sum y}{150} = \dfrac{150}{150} = 1$

$\bar{y} = \bar{x} - 154 \Rightarrow \bar{x} = \bar{y} + 154 = 155$

An unbiased estimate of population mean is $\bar{x} = 155$.

> Recall:
> If $y = ax + b$, then
> $\bar{y} = a\bar{x} + b$ and $s_y{}^2 = a^2 s_x{}^2$.

$$s_y{}^2 = \frac{1}{n-1}\left(\sum y^2 - \frac{\left(\sum y\right)^2}{n}\right)$$

$$= \frac{1}{149}\left(11000 - \frac{(150)^2}{150}\right)$$

$$= \frac{10850}{149} = 72.8 \text{ (3 sf).}$$

$s_y{}^2 = s_x{}^2$

An unbiased estimate of population variance is $s_x{}^2 = \dfrac{10850}{149}$ or 72.8 (3 sf).

3 The amount, x mg, of Vitamin B2 in a packet of cereals was measured. 10 packets of cereals were taken and the following data were obtained:

272, 285, 278, 293, 298, 283, 279, 281, 295, 271

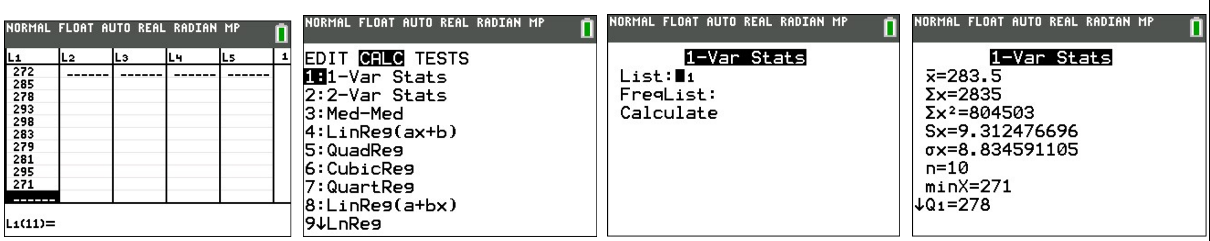
Find unbiased estimates of the population mean and population variance of Vitamin B2 in packets of the cereal.

[283.5, 86.7]

Solution:

Using GC,
an unbiased estimate of the population mean of Vitamin B2 in a packet of cereal is 283.5 mg.

An unbiased estimate of the population variance of Vitamin B2 in a packet of cereal is
$s^2 \approx (9.3125)^2 \text{mg}^2 = 86.7 \text{ mg}^2$ (3 s.f.)



4 Given that the random variables X and Y have distributions $\mathrm{N}(30,18)$ and $\mathrm{N}(20,16)$ respectively, and that $\overline{X}$ and $\overline{Y}$ denote the means of 15 independent observations of X and 8 independent observations of Y respectively.

State the sampling distributions of

(i) $\overline{X}-\overline{Y}$,

(ii) $5\overline{X}+3\overline{Y}$,

(iii) $\dfrac{X_1+X_2+\ldots+X_{15}+Y_1+Y_2+\ldots+Y_8}{23}$.

Solution:	
Given $X\sim\mathrm{N}(30,18)$ and $Y\sim\mathrm{N}(20,16)$ with $n_x=15$ and $n_y=8$ Then, $\overline{X}\sim\mathrm{N}\left(30,\ \dfrac{18}{15}\right)$ and $\overline{Y}\sim\mathrm{N}\left(20,\ \dfrac{16}{8}\right)$ i.e., $\overline{X}\sim\mathrm{N}(30,\ 1.2)$ and $\overline{Y}\sim\mathrm{N}(20,\ 2)$ Since X and Y follow normal distributions, the sample means ie $\overline{X}$ and $\overline{Y}$ follow normal distributions too.	
(i)	$\mathrm{E}\left(\overline{X}-\overline{Y}\right)=\mathrm{E}\left(\overline{X}\right)-\mathrm{E}\left(\overline{Y}\right)=30-20=10$ $\mathrm{Var}\left(\overline{X}-\overline{Y}\right)=\mathrm{Var}\left(\overline{X}\right)+\mathrm{Var}\left(\overline{Y}\right)=1.2+2=3.2$ $\overline{X}-\overline{Y}\sim\mathrm{N}(10,\ 3.2)$
(ii)	$\mathrm{E}\left(5\overline{X}+3\overline{Y}\right)=5\mathrm{E}\left(\overline{X}\right)+3\mathrm{E}\left(\overline{Y}\right)=5(30)+3(20)=210$ $\mathrm{Var}\left(5\overline{X}+3\overline{Y}\right)=5^2\,\mathrm{Var}\left(\overline{X}\right)+3^2\mathrm{Var}\left(\overline{Y}\right)=5^2(1.2)+3^2(2)=48$ $5\overline{X}+3\overline{Y}\sim\mathrm{N}(210,\ 48)$
(iii)	Let $T=\dfrac{X_1+X_2+\ldots+X_{15}+Y_1+Y_2+\ldots+Y_8}{23}$ $\mathrm{E}(T)=\dfrac{1}{23}\left[15\mathrm{E}(X)+8\mathrm{E}(Y)\right]=\dfrac{610}{23}$ $\mathrm{Var}(T)=\left(\dfrac{1}{23}\right)^2\left[15\mathrm{Var}(X)+8\mathrm{Var}(Y)\right]=\dfrac{398}{529}$ $T\sim\mathrm{N}\left(\dfrac{610}{23},\dfrac{398}{529}\right)$

5 A random variable X has expectation 1 and variance $\frac{1}{3}$. The random variable S is the sum of 80 independent observations of X.

Find an approximate value for $P(75 < S < 90)$.

[0.807]

9233/2002/02/Q30 OR (modified)

Solution:

Note:
We do not have information on the distribution of X.

Given $E(X)=1$, $\text{Var}(X)=\frac{1}{3}$ and $S = X_1 + X_2 + ... + X_{80}$

Since $n = 80$ is large, by Central Limit Theorem,

$$S \sim N\left(80(1),\ 80\left(\frac{1}{3}\right)\right) \text{ approximately.}$$

X is not known to follow a normal distribution.
However, since n is large, by Central Limit Theorem, $S = X_1 + X_2 + ... + X_{80}$ follows a normal distribution *approximately*.

$$\text{i.e } S \sim N\left(80,\ \frac{80}{3}\right)$$

$P(75 < S < 90) = 0.807$ (3 s.f.)

OR

Given $E(X)=1$, $\text{Var}(X)=\frac{1}{3}$ and $\bar{X} = \frac{X_1 + X_2 + ... + X_{80}}{80}$

Since $n = 80$ is large, by Central Limit Theorem,

$$\bar{X} \sim N\left(1,\ \frac{1/3}{80}\right) \text{ approximately.}$$

X is not known to follow a normal distribution.
However, since n is large, by Central Limit Theorem, the sample mean $\bar{X}$ follows a normal distribution *approximately*.

$$\text{i.e } \bar{X} \sim N\left(1,\ \frac{1}{240}\right)$$

$P(75 < S < 90) = P\left(\frac{75}{80} < \bar{X} < \frac{90}{80}\right) = 0.807$ (3 s.f.)